

llmXive Automated Review of MemGUI-Agent: An End-to-End Long-Horizon Mobile GUI Agent with Proactive Context Management

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_claim_accuracy.md

The paper presents several strong factual claims regarding performance gains and architectural impacts. Most claims are supported by the provided tables, but a few require clarification to ensure the evidence matches the strength of the assertions.

In Section 3.1, the claim that smaller models “regress” with CONACT is supported by Table 1, which shows a drop in Pass@1 for 2B, 4B, and 8B models. However, the text does not explicitly distinguish this as a zero-shot phenomenon, which is crucial because the subsequent section (3.2) introduces MemGUI-3K specifically to fix this. Without this distinction, a reader might incorrectly infer that the CONACT architecture is detrimental to smaller models in general, rather than just ineffective without the specific SFT data. The text should clarify that the regression is observed in the zero-shot setting and is reversed by the proposed SFT.

In Section 4.1, the claim that MemGUI-8B-SFT achieves the “best open-data 8B performance” is supported by Table 2, which compares it against Qwen3-VL-8B-Instruct. However, the Related Work section and the bibliography list other potential open-data 8B models (e.g., UI-Venus, Mobile-Agent-V3). The review cannot verify if these models were excluded from the comparison or if they simply performed worse. To support the “best among open-data” claim robustly, the authors should either include these models in the comparison table or explicitly state in the text that the comparison is limited to the specific baselines chosen and that other open-data models were considered but not included for specific reasons (e.g., lack of public weights, different benchmark settings).

In Section 4.3, the ablation study text states that “history folding adds 5.0%, and self-describing steps add 7.5%”. The table shows these as cumulative improvements in a sequential ablation (Baseline -> +UI -> +History -> +Self). The phrasing “adds X%” could be interpreted as an independent contribution to the baseline, which is not what the table shows. The text should be refined to clarify that these are the marginal gains observed when adding each component to the *previous* configuration in the ablation sequence.

Finally, in Section 4.4, the claim about reducing failures by 41% is mathematically sound based on the numbers provided (99 to 58). The breakdown of reductions in process and output hallucination (-42% and -57%) is likely correct based on the figure, but the text should ensure it is clear that these percentages refer to the reduction in the *count* of those specific error types, not their contribution to the total reduction, to avoid any ambiguity in how the total 41% reduction is composed.

Overall, the claims are largely accurate, but minor clarifications in the text are needed to prevent misinterpretation of the data, particularly regarding the zero-shot vs. SFT performance of smaller models and the specific nature of the ablation gains.

Action items.

- **[writing]** Clarify in Section 3.1 that the performance regression for smaller models (2B-8B) with CONACT is specific to the zero-shot setting, and that SFT (MemGUI-8B-SFT) reverses this trend, to avoid implying the architecture is inherently harmful.
- **[science]** In Section 4.1, verify that ‘Qwen3-VL-8B-Instruct’ is the only relevant open-data 8B baseline omitted from Table 2. If other open-data 8B models exist (e.g., UI-Venus), either include them or qualify the ‘best among open-data’ claim to reflect the specific baselines compared.
- **[writing]** In Section 4.3, clarify that the reported gains for ‘history folding’ (+5.0%) and ‘self-describing steps’ (+7.5%) are marginal improvements in the sequential ablation, not independent additive contributions to the ReAct baseline.
- **[writing]** In Section 4.4, explicitly state that the -42% and -57% reductions for hallucination types refer to the decrease in the count of those specific errors, not their proportional contribution to the total 41% failure reduction, to prevent ambiguity.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

The figure provides a clear visual overview of the data collection pipeline, but the final statistics box contains a typo (‘stpes’) and an undefined symbol is used in the agent teacher section.

Figure 2

The figure effectively illustrates a training trajectory with reasonable and unreasonable steps, but the visual labeling of ‘reasonable’ steps is implicit rather than explicit, and the top instruction text is partially truncated.

Figure 3

The figure effectively displays the training dynamics with clear trends, but the x-axis labels are illegible watermarks, and the perfectly linear step/epoch plots appear synthetic rather than empirical.

Figure 4

The figure effectively illustrates a specific failure trajectory with clear visual annotations, but the caption is too brief to support the detailed content shown, failing to describe the task, the failure mode, or the evaluation result.

Figure 5

The figure presents a raw failure log and decision block rather than a structured visualization of the ReAct-style prompting process described in the caption, creating a disconnect between the visual evidence and the textual claim.

Figure 6

The figure displays a complex failure trajectory with detailed annotations, but the caption is insufficient, failing to describe the task, the specific failure modes illustrated, or the meaning of the visual annotations.

Figure 7

The figure is a failure case visualization that completely mismatches its caption and filename, which suggest a benchmark result or chart. The image lacks a proper title or legend to explain the specific task or failure mode depicted.

Figure 8

Figure 8 effectively illustrates a representative process-hallucination failure by displaying a chronological sequence of 16 mobile GUI screenshots annotated with the agent’s actions and internal reasoning. The visual narrative clearly demonstrates the agent deviating from the required workflow and falsely assuming task completion, which aligns perfectly with the provided caption.

Figure 9

Figure 9 effectively illustrates an output-hallucination failure by displaying a chronological sequence of mobile UI screenshots. The figure clearly highlights the agent’s inability to find specific data (Autel EVO Lite+ price/rating) and its subsequent fabrication of this information in the final SMS message, which aligns perfectly with the provided caption.

Figure 10

The figure effectively illustrates a knowledge-deficiency failure with detailed step-by-step screenshots and annotations. However, the caption is incomplete as it fails to describe the prominent instruction and evaluation blocks or the meaning of the red annotation callouts.

Figure 11

Figure 11 effectively illustrates an intent-understanding failure by presenting a chronological sequence of screenshots with clear annotations. The visual evidence of the agent’s actions, combined with the explanatory text bubbles and the summary at the bottom, aligns perfectly with the caption’s description of the error.

Figure 12

The figure effectively visualizes successful agent trajectories with clear screenshots and text snippets, but the caption contains unrendered template placeholders and slightly confusing directional instructions.

Action items.

- **[writing]** Figure 1: The final output label ‘MemGUI-3K’ lists ‘64,430 stpes’ with a typo (‘stpes’ instead of ‘steps’).
- **[writing]** Figure 1: The ‘Five-part ConAct Rollout’ box contains a symbol (interlocking rings) that is not defined in the figure or caption.
- **[science]** Figure 2: The caption states the trajectory contains ‘step-level reasonableness labels,’ but the visual only labels steps 8 and 9 as ‘Unreasonable’ in a summary box at the bottom; individual step cards (e.g., step 1) lack explicit ‘Reasonable’ labels, making the annotation scheme visually ambiguous.
- **[writing]** Figure 2: The instruction text at the top is cut off on the right side (‘...published more recently.’ is incomplete in the screenshot), reducing readability.
- **[writing]** Figure 3: The x-axis label ‘train/global_step’ is rendered as a tiny, illegible watermark in the bottom-right corner of the subplots rather than a clear axis label.
- **[writing]** Figure 3: The y-axis for ‘train/learning_rate’ lacks a unit (e.g., ‘lr’ or ‘1e-4’), making the absolute scale ambiguous without external context.
- **[science]** Figure 3: The ‘train/global_step’ and ‘train/epoch’ subplots display perfectly linear, noise-free lines, which is atypical for training logs and suggests these are synthetic or idealized plots rather than empirical data.
- **[science]** Figure 4: The figure displays a complex multi-step failure trajectory with specific annotations (red circles, text bubbles) and a ‘MemGUI-Eval’ decision block, but the caption ‘Qwen3-VL-8B-Instruct base’ is insufficient to explain the content, context, or the specific failure mode being illustrated.
- **[writing]** Figure 4: The figure title ‘Trajectory of Qwen3-VL 8B-Instruct’s Failure in MemGUI-Bench’ and the bottom ‘MemGUI-Eval’ block are not referenced or described in the provided caption, creating a disconnect between the visual evidence and the figure description.
- **[science]** Figure 5: The caption claims to show ‘Qwen3-VL-235B-Thinking with ReAct-style prompting,’ but the image displays a ‘Failed’ decision with a ‘Reason’ block explaining the

failure. This contradicts the caption’s implication of a successful demonstration or standard ReAct output, and the figure lacks the specific ‘Thought/Action/Observation’ formatting typically associated with ReAct style, instead showing a raw failure log.

- **[writing]** Figure 5: The image is a screenshot of a failure case (labeled ‘MemGUI-Eval’ and ‘Failed’) rather than a structured visualization of the ‘ReAct-style prompting’ process. It is unclear how this specific failure trajectory illustrates the prompting method described in the caption.
- **[science]** Figure 6: The rendered image is a detailed 20-step failure trajectory of a mobile agent, but the caption describes it merely as ‘Qwen3-VL-8B-Instruct base’ without explaining the content, task, or failure mode shown.
- **[science]** Figure 6: The image contains specific annotations (e.g., ‘Stops after one swipe’, ‘Hallucinates fake number’) and a final ‘Failed’ decision, but the caption provides no context for these elements or the task being performed.
- **[fatal]** Figure 7: The rendered image is a detailed failure case walkthrough of a mobile agent task (labeled ‘Trajectory of Qwen3-VL 235B-Thinking’s Failure in MobileWorld’), which contradicts the caption ‘Qwen3-VL-235B-Thinking with ReAct-style prompting’ and the filename ‘MobileWorld-base-235b.png’ (which implies a benchmark result table or chart). The image content does not match the figure label.
- **[fatal]** Figure 7: The image contains a large text overlay at the top (‘My friend Olivia has left...’) and a bottom overlay (‘<Decision> Failed’) that are not defined in the caption, making the figure’s purpose ambiguous without external context.
- **[writing]** Figure 10: The image contains a large ‘Instruction’ block at the top and a ‘MemGUI-Eval’ decision block at the bottom, which are not mentioned in the caption. The caption should explicitly describe these components to clarify the figure’s structure.
- **[writing]** Figure 10: The red annotations (e.g., ‘Directly adding ingredients after the title line...’) are not defined in the caption. The caption should explain that these callouts highlight specific UI operation failures.
- **[writing]** Figure 12: The caption contains unrendered variable placeholders ‘casebenchpurplebg’ and ‘casebenchbluebg’ instead of the actual dataset names (MemGUI-Bench and MobileWorld), likely due to a template rendering error.
- **[writing]** Figure 12: The caption states ‘Read rows left to right’, but the layout consists of two distinct horizontal panels (one for each task) rather than a single multi-row table, making the directional instruction confusing.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript relies heavily on domain-specific acronyms and jargon that are not defined upon first use, creating barriers for non-specialist readers.

In the **Introduction (Section 1)**, the term **ReAct** is used immediately without expansion. While standard in the sub-field, it should be spelled out as “Reasoning and Acting” on first mention. Similarly, **SFT** (Supervised Fine-Tuning) appears in Section 1 and Section 3 without definition. The phrase “prompt explosion” is used metaphorically; replacing this with “excessive prompt length” or “context bloat” would be more precise and accessible.

In **Section 3 (Dataset)**, the term **backbone** is used to refer to the base model. While common in deep learning, “base model” or “foundation model” is clearer for a broader audience. The term **rollouts** is used to describe teacher-generated trajectories; “simulated runs” or “generated trajectories” is less jargon-heavy.

In **Section 4 (Experiments)** and the **Appendix**, several critical metrics are introduced without definition. **IRR** (Information Retention Rate) and **MTPR** (Memory-Task Proficiency Ratio) appear in tables (Table 1, Table 3) but are not defined in the main text. **Pass@k** is used extensively but lacks a formal definition in the body. **LoRA** is mentioned in the appendix without expansion.

Finally, the term **zero-shot** is used frequently (e.g., Section 1, Section 3) without a brief parenthetical explanation (e.g., “without task-specific fine-tuning”), which would aid readers unfamiliar with LLM evaluation protocols.

These omissions do not invalidate the science but significantly reduce the paper’s accessibility to the broader computer science community and practitioners outside the specific niche of LLM agents.

Action items.

- **[writing]** The manuscript relies heavily on domain-specific acronyms and jargon that are not defined upon first use, creating barriers for non-specialist readers. In the Introduction (Section 1), the term ReAct is used immediately without expansion. While standard in the sub-field, it should be spelled out as “Reasoning and Acting” on first mention. Similarly, SFT (Supervised Fine-Tuning) appears in Section 1 and Section 3 without definition. The phrase “prompt explosion” is used metaphorically; replacing

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a coherent argument that proactive context management (CONACT) solves the “prompt explosion” issue in long-horizon mobile GUI agents. The logical flow from the

problem statement (Section 1) to the proposed mechanism (Section 2) and empirical validation (Section 5) is generally sound. However, there are specific gaps in the logical consistency between the stated claims and the provided evidence.

First, in Section 3.1, the authors motivate the need for the MemGUI-3K dataset by stating that “Zero-shot CONACT only benefits the strongest backbone” and that “smaller models regress.” While Table 1 confirms regression for the 2B, 4B, and 8B models, it also shows a regression for the Qwen3-VL-235B-Instruct model (23.4% to 19.5% Pass@1). The text concludes that “Only 235B-Thinking benefits,” but logically, the failure of the 235B-Instruct variant (which is of similar scale) suggests that model size alone is not the sole differentiator. The paper lacks a logical explanation for why the “Thinking” variant succeeds where the “Instruct” variant fails, weakening the causal claim that the method is universally applicable to large models without specific architectural or training nuances.

Second, the error analysis in Section 5.1 claims that full CONACT reduces “process hallucination (-42%)” and “output hallucination (-57%).” The text references Figure 2 (failure_heatmap) as the source. However, the figure caption provided in the source only reports the aggregate reduction in total failures (99 to 58, a 41% drop). The specific percentages for the sub-categories of hallucination are not present in the text or the figure description. This creates a logical gap where the specific quantitative claims cannot be verified by the cited evidence.

Finally, the ablation study in Table 3 presents a logical inconsistency regarding component synergy. The text and table show that adding UI memory actions (+12.5%), history folding (+5.0%), and self-describing steps (+7.5%) to the baseline (5.0%) results in a cumulative sum of 25.0%. However, the “Full CONACT” configuration achieves 40.0% Pass@1. The paper does not logically account for the missing 15.0% gain. Without an explanation of the non-linear interactions or synergistic effects between these components, the claim that the full system’s performance is a direct result of these additive parts is unsupported. The mechanism by which the components interact to produce a super-additive effect is missing.

Action items.

- **[science]** In Section 3.1 (Table 1), the text claims smaller models ‘regress’ with CONACT, yet the table shows Qwen3-VL-235B-Instruct also regresses (23.4% -> 19.5% P@1). The conclusion that ‘Only 235B-Thinking benefits’ is logically incomplete without explaining why the Instruct variant fails despite similar scale.
- **[writing]** Section 5.1 claims CONACT reduces ‘process hallucination (-42%)’ and ‘output hallucination (-57%)’. The text cites Figure 2 (failure_heatmap) for this, but the figure caption only lists total failure counts (99->58). The specific breakdown percentages for these two categories are missing from the text and cannot be verified from the provided figure description.
- **[science]** The ablation study (Table 3) presents additive gains for components (+12.5, +5.0, +7.5) summing to 25.0, yet the full system achieves 40.0. The text does not explain the non-linear synergy or interaction effects that account for the remaining 15.0% gain, leaving the causal mechanism of the full system’s performance unexplained.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several strong claims regarding the efficacy of the proposed CONACT framework and the MemGUI-3K dataset. While the experimental results generally support the core hypothesis that proactive context management improves long-horizon performance, there are instances of over-claiming where the language extrapolates beyond the specific conditions of the reported data.

First, the Introduction states that CONACT “saves ~1.5k input tokens by step 150 compared to ReAct.” This is a precise quantitative claim that currently lacks the necessary context to be verified. The paper does not explicitly define the baseline ReAct context growth rate (tokens per step) or the specific compression ratio achieved by the folding mechanism that leads to this 1.5k figure. Without a clear definition of the baseline or a reference to a specific calculation in the appendix, this number appears to be an extrapolation rather than a directly measured result in the provided text. The authors should clarify the baseline assumptions or provide the calculation methodology.

Second, the Conclusion asserts that “MemGUI-8B-SFT generalizes to MobileWorld.” While the model does show improvement over its zero-shot baseline on MobileWorld (Table 4), the term “generalizes” in this context risks overstating the capability. The achieved Success Rate (17.9%) is significantly lower than other agents on the same benchmark (e.g., GUI-Owl-1.5-32B at 43.9%). The claim should be tempered to reflect that the model demonstrates *relative* improvement or *transfer* of the CONACT strategy to a new benchmark, rather than implying a robust, general-purpose capability that matches or exceeds existing specialized agents.

Finally, the error analysis in Section 5.4 claims that “Full CONACT reduces total failures by 41% (99→58).” This statistic is derived from a specific comparison between the Full CONACT variant and the ReAct baseline using the Qwen3-VL-235B-Thinking model. The phrasing could be misinterpreted as a universal property of the method across all model scales or task difficulties. Given that Table 1 shows smaller models (2B, 4B, 8B) actually regress when using CONACT zero-shot, the 41% reduction is not a generalizable fact for the method itself but rather a result specific to the 235B model configuration. The text should explicitly qualify this reduction to the specific experimental setup to avoid misleading readers about the method’s universal applicability.

Action items.

- **[writing]** The claim that CONACT ‘saves ~1.5k input tokens by step 150’ (Intro) lacks a defined baseline comparison or calculation methodology. Specify the exact ReAct context growth rate and the folding compression ratio used to derive this figure.
- **[writing]** The statement that MemGUI-8B-SFT ‘generalizes to MobileWorld’ (Conclusion) overstates the evidence. The results in Table 4 show a 17.9% SR, which is lower than the

43.9% SR of GUI-Owl-1.5-32B. Clarify that the gain is relative to the 8B baseline, not a generalization to state-of-the-art performance.

- **[writing]** The claim that the method ‘reduces total failures by 41%’ (Sec 5.4) is based on a specific ablation variant (Full CONACT vs ReAct) on a specific model (235B). Ensure the text does not imply this reduction applies universally to all model sizes or task difficulties without qualification.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

Safety and Ethics Review

The manuscript presents MemGUI-Agent, an autonomous mobile GUI agent capable of long-horizon task execution. While the technical contribution is significant, the paper currently lacks sufficient detail regarding the ethical implications of the data used and the potential risks of deploying such an autonomous agent.

1. Data Provenance and Privacy (Section 3) The construction of the **MemGUI-3K** dataset is central to the paper’s claims. Section 3.2 mentions expanding 128 seed tasks via “entity substitution” and “task simplification.” However, the source of these seed tasks is not explicitly defined.

- If the seed tasks were derived from real user interactions or scraped from public app stores/forums, the authors must address **data privacy**. Specifically, a statement is needed confirming that all Personally Identifiable Information (PII) (e.g., user names, phone numbers, addresses, financial data) was scrubbed from the trajectories before inclusion in the dataset.
- If human annotators were involved in creating or verifying the trajectories, the paper must state whether this process received **IRB (Institutional Review Board)** approval or adhered to standard ethical guidelines for human-subject research. Currently, Section 3.2 and the Appendix (Section 4) lack this declaration.

2. Dual-Use and Autonomous Action Risks (Section 2 & 6) The agent is designed to execute actions (**click**, **type**, **swipe**) autonomously on a mobile device. This capability introduces **dual-use risks**:

- **Malicious Automation:** The agent could theoretically be repurposed to perform unauthorized actions, such as making in-app purchases, sending messages to contacts, or scraping sensitive data from a device without the user’s explicit, real-time consent.
- **Lack of Safety Guardrails:** The current prompt templates (Appendix) and action space do not appear to include explicit “safety checks” or “human-in-the-loop” confirmation steps for high-risk actions (e.g., deleting data, transferring money, or changing system settings).

- **Recommendation:** The authors should add a dedicated paragraph in the **Limitations** section (Section 5) or a new **Safety Considerations** section. This should explicitly state that the agent is intended for benign research purposes, acknowledge the potential for misuse, and clarify that the current implementation does not include built-in safeguards against malicious deployment.

3. Benchmark Integrity The benchmarks (MemGUI-Bench, MobileWorld) involve interacting with real or simulated apps. The authors should confirm that the evaluation protocol does not involve interacting with live user accounts or sensitive data during the automated testing phase, ensuring that the benchmarking process itself does not violate privacy norms or terms of service of the applications tested.

Conclusion The paper is technically sound but requires minor revisions to address ethical transparency regarding data collection and to explicitly acknowledge the safety risks associated with autonomous mobile agents. Adding the requested statements will ensure the work meets standard ethical publication requirements.

Action items.

- **[writing]** The paper lacks an explicit statement regarding IRB approval or ethical review for the human-in-the-loop data collection or annotation processes used to construct the MemGUI-3K dataset. Please add a statement in Section 3 (Dataset) confirming whether human annotators were involved and if their participation was reviewed by an ethics board.
- **[writing]** The dataset construction involves ‘entity substitution’ and ‘task simplification’ on seed tasks. The authors must clarify the provenance of these seed tasks. If they were scraped from public app stores or user forums, a statement on data privacy, terms of service compliance, and the exclusion of PII (Personally Identifiable Information) is required.
- **[writing]** The agent is designed to perform autonomous actions (click, type, swipe) on mobile devices. The paper should include a ‘Safety and Limitations’ subsection explicitly stating that the agent is not intended for use in environments where it could cause financial harm, privacy violations, or unauthorized access, and that it lacks a ‘human-in-the-loop’ safety override mechanism.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The scientific evidence supporting the core claims of MemGUI-Agent is generally strong, particularly in the magnitude of performance gains observed on long-horizon tasks. The ablation study (Table 3) effectively isolates the contribution of the CONACT components, showing a substantial jump from 5.0% to 40.0% Pass@1 when adding UI memory actions and folding. The dataset construction (MemGUI-3K) appears rigorous, with a clear pipeline for expanding seed

tasks and filtering teacher rollouts.

However, several aspects of the evidence require strengthening to rule out alternative explanations and ensure statistical robustness. First, the ablation results in Table 3 are presented as point estimates without measures of variance or statistical significance. With a test set of only 128 tasks (MemGUI-Bench), a 35% absolute gain is large, but without p-values or confidence intervals (e.g., via bootstrapping), it is difficult to rule out that the gains are due to random variance or specific task selection rather than the architectural change. The authors should report statistical significance for the key ablation comparisons.

Second, the “reasonableness filtering” step in the dataset construction (Section 3.2, Appendix D.3) is a potential source of selection bias. The paper states that teacher rollouts are filtered for “reasonableness” but does not define the criteria or report the rejection rate. If the filter disproportionately removes difficult or ambiguous trajectories, the resulting SFT dataset may be biased toward easier tasks, artificially inflating the performance of MemGUI-8B-SFT. The authors should clarify the filtering criteria and report the percentage of trajectories/steps rejected at this stage.

Finally, the claim that MemGUI-8B-SFT is the “best open-data 8B model” (Section 4.1) is currently supported only by comparisons against the base Qwen3-VL-8B-Instruct and a few proprietary frameworks. To substantiate this claim, the authors should include comparisons against other recent open-weight 8B models that have been fine-tuned for GUI tasks (e.g., UI-TARS, Mobile-Agent-V3 variants) on the same benchmarks. Without these baselines, the “best” claim remains relative to a limited set of competitors.

Action items.

- **[science]** The ablation study (Table 3) presents component contributions (e.g., +35.0% P@1 for UI memory) but lacks statistical significance testing (e.g., p-values or confidence intervals). Given the small test set size (128 tasks), random variance could explain these gains. Please report significance tests or bootstrap CIs.
- **[science]** The dataset construction relies on ‘reasonableness filtering’ of teacher rollouts (Sec 3.2, Appx D.3). The criteria for ‘reasonable’ are not quantified, and the filter rate is not reported. This introduces potential selection bias where only easy or obvious trajectories are retained, inflating SFT performance. Clarify the filtering protocol and report the rejection rate.
- **[science]** The claim that MemGUI-8B-SFT is the ‘best open-data 8B model’ (Sec 4.1) is based on a single benchmark (MemGUI-Bench) and MobileWorld. No comparison is made against other open-weight 8B models fine-tuned on similar data (e.g., UI-TARS, Mobile-Agent-V3 variants) on these specific long-horizon metrics. Broaden the baseline comparison to ensure the ‘best’ claim is robust.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical analysis presented in the manuscript is largely descriptive, relying on point estimates (percentages and means) without accompanying measures of uncertainty or significance testing. While the magnitude of the reported improvements (e.g., +13.3 Pass@1 in Table 1) is substantial, the absence of confidence intervals or p-values makes it difficult to assess the robustness of these gains against the inherent variance in LLM-based agent evaluation.

Specifically, in **Table 1 (Main Results)** and **Table 2 (Ablation Study)**, the authors compare multiple models and ablation variants. Standard practice for such benchmarks requires reporting the standard deviation over multiple runs or, at minimum, bootstrapped confidence intervals to demonstrate that the observed differences are not due to random fluctuations in the test set or stochasticity in the model generation. The current text states “significantly improves” in Section 4.2 regarding the offline metrics, but no statistical test (e.g., t-test, Wilcoxon signed-rank) is cited to support this claim.

Furthermore, the **Offline Skill Analysis** (Table 3) presents metrics like “Memory Trigger F1” and “Deep Fold Ratio” derived from the MemGUI-3K test set. The manuscript does not specify the sample size (number of steps or trajectories) used to compute these aggregates, nor does it provide error bars. Given that the dataset is constructed via teacher rollouts and filtering, understanding the variance in these metrics is crucial for reproducibility.

Finally, in **Section 3.3**, the average trajectory length is reported as 28.8 steps. Without the standard deviation or a histogram (which is referenced but not visible in the text source), the distribution of task difficulty remains opaque. A narrow distribution could skew results, while a wide one might mask performance drops on specific sub-populations of tasks. The authors should supplement their point estimates with measures of dispersion and statistical significance tests to strengthen the empirical claims.

Action items.

- **[science]** Report statistical significance (e.g., p-values, confidence intervals, or bootstrapped error bars) for the performance gains in Table 1 (Main Results) and Table 2 (Ablation). The current presentation of point estimates (e.g., 37.5% vs 32.8%) lacks evidence of robustness against variance in the evaluation protocol.
- **[science]** Clarify the statistical methodology for the ‘Offline Skill Analysis’ (Table 3). Specifically, define the sample size (N) used for the F1 and accuracy metrics and state whether the reported improvements (e.g., 19.9% to 48.0%) are statistically significant or if they represent mean values over a single test split.
- **[writing]** In Section 3.3 (Dataset Statistics) and Appendix D, specify the variance or standard deviation for the reported trajectory lengths (avg. 28.8 steps) and fold spans. A single mean value is insufficient to characterize the distribution of task complexity used for training and evaluation.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript demonstrates a high level of technical clarity and logical flow, effectively communicating the motivation for proactive context management in mobile GUI agents. The introduction successfully sets up the problem of “prompt explosion” and the proposed solution, \conact, with clear definitions of the three structured fields. The writing is generally concise, and the transition from the problem statement to the methodology is smooth.

However, there are minor issues regarding the presentation of technical details and sentence-level precision that should be addressed before final publication.

First, in Section 2 (“Method”), the description of the “Step output protocol” includes a `Verbatim` environment that contains raw LaTeX escape sequences (e.g., `\{\}`) intended to represent JSON-like structures. While the intent is clear, this formatting choice risks rendering issues in standard PDF viewers or LaTeX compilers that do not handle these specific escape sequences within `Verbatim` blocks gracefully. It is recommended to replace this with a standard `lstlisting` environment or a `verbatim` block with properly escaped characters to ensure the code snippet is readable and compiles without warnings.

Second, in Section 3.1 (“Motivation”), the sentence “Zero-shot \conact only benefits the strongest backbone... while smaller models regress” is slightly ambiguous without immediate context. While Table 1 clarifies this, the text itself would benefit from a brief clarification that the regression is relative to the ReAct baseline. Adding a phrase such as “regress relative to the ReAct baseline” would make the claim self-contained and improve readability for a reader skimming the text.

Finally, the caption for Figure 1 contains a minor redundancy. The phrase “Context management in mobile GUI agents” appears in the caption setup (`\captionsetup`) and is repeated verbatim at the start of the caption body. Removing the repetition in the body text will streamline the presentation.

Overall, the paper is well-written and the narrative is compelling. Addressing these minor formatting and phrasing points will polish the manuscript to a publication-ready standard.

Action items.

- **[writing]** In Section 2, the phrase ‘The model emits a 5-part structured output’ is followed by a `Verbatim` block that uses non-standard LaTeX syntax (e.g., `\{\}`) which may not render correctly in all viewers. Ensure the code block is properly escaped or use a standard `lstlisting` environment for better readability and compilation safety.
- **[writing]** Section 3.1 (Motivation) states ‘smaller models regress’ but does not explicitly define the baseline for regression in the text (though Table 1 shows it). Clarify that regression is relative to the ReAct baseline to ensure the sentence stands alone without requiring the reader to cross-reference the table immediately.

- **[writing]** The caption for Figure 1 (teaser) is repetitive: ‘Context management in mobile GUI agents.’ appears twice (once in the caption text, once in the caption setup). Remove the redundant phrase in the caption body to improve flow.